Ders 4

Eksoru: Why smith chart is used · (List 6)

Eksoru: Write S measurement Devices

Eksoru: Write the impedan the real (P) and Complex Q power for Resistor, Inductor Capacitor

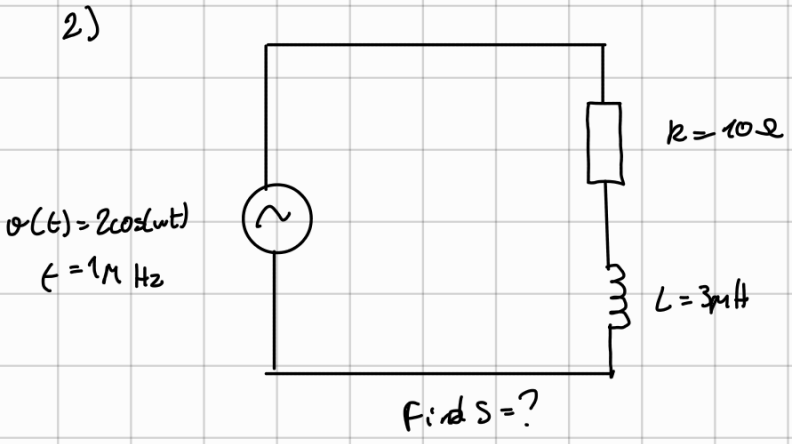
HW ①

1) $v(t) = 2\cos(\omega t + 30)$

$Z = 5 + j10$

Find I and Pave

2)



What is the drawbacks of Nonlinearity :?

Ders5

Ex



Vericinin ikinci harmoniğine kadar bastırılmalıdır ki bu alıcı doğurma ihtimali

Eksorm

$IM3 = ?$ $A_{IIP3} = ?$ $A_{1dB} = ?$

$A^2_{IIP3} = ?$

Ex:

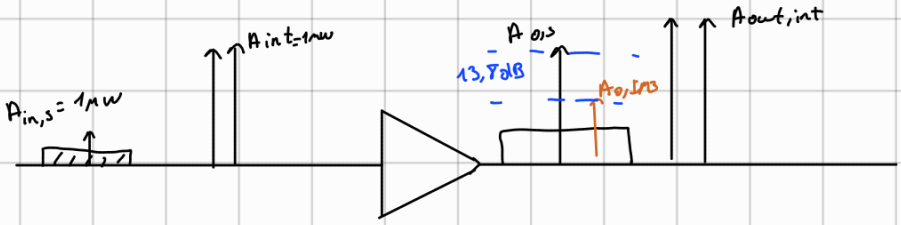
$A_{in,s} = 1\mu V$

$A_{in,int} = 1 mV$

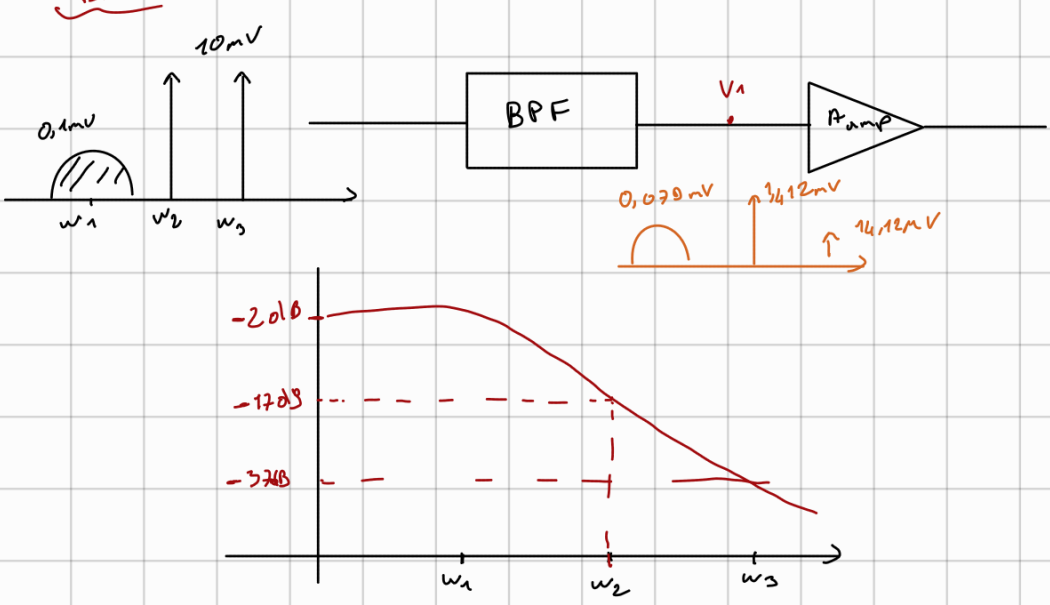
$P_{IIP3} = -10\ dBm$

$Z = 50\ \Omega$

Find $\frac{A_{o,s}}{A_{o,IM3}} = ?$



Ex



Compute IIP3 of the amplifier such that IM3 product falling at ω_1 is 20 dB below the desired signal ?

→ $A_{o,IM3}$??

↳ $A_{o,s}$

Ex:

Çalışma soruları

$a_1 = 10$

$A_{in,int} = 10\,mV$

$A_{IIP3} = 70\,mV$

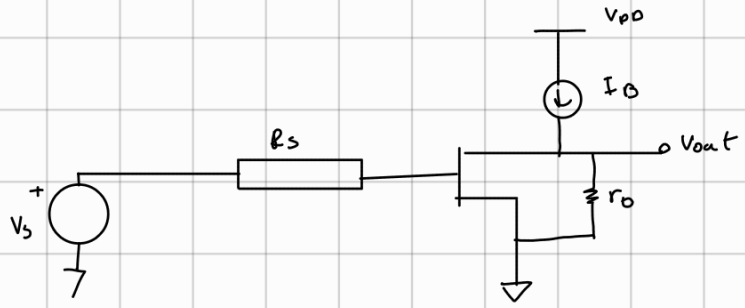
} IM3=?, $A_{o,IM3}$ = ?

Draw resistor and Mos Noise model:

Ders 6

How to find input referred Noise

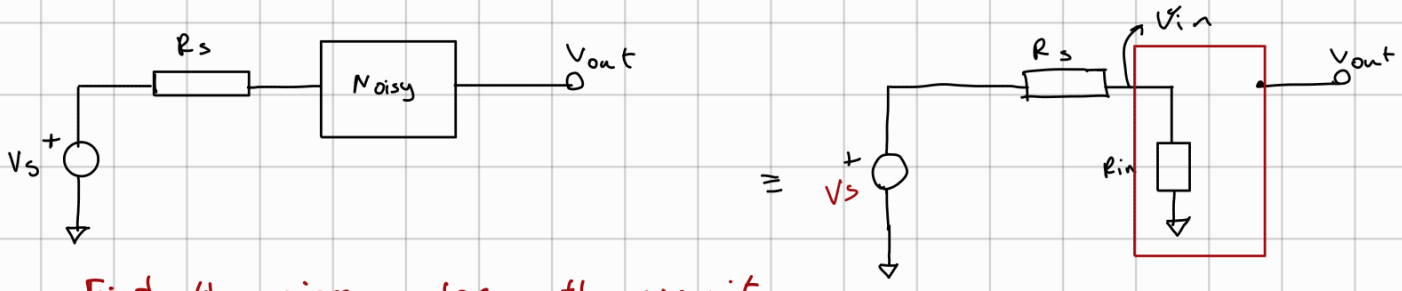
Ex



$$\overline{v_{n,out1}^2} + \overline{v_{n,out2}^2} + \overline{v_{n,out3}^2} + \cdots = \overline{v_{n,out}^2}_{(total)}$$

$$\overline{v_{n,input}^2} = \frac{\overline{v_{n,out\,(total)}^2}}{A_{vt}^2}$$

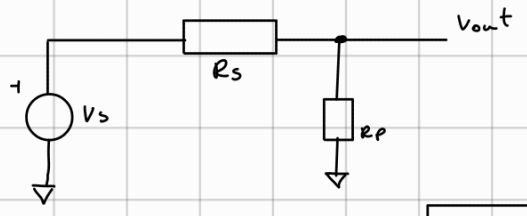
Noise Figure



Find the noise factor of the circuit

Hint : $SNR_{in} = \frac{P_{s,in}}{P_{n,in}}$ $SN_{out} = \frac{P_{s,out}}{P_{nout}}$ $F = \frac{SNR_{in}}{SNR_{out}}$

Ex:



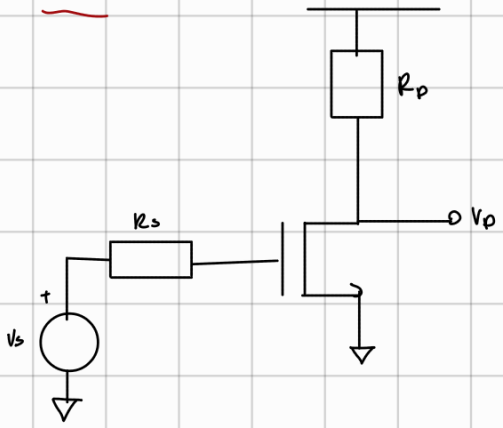
$$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{4kTR_s}$$,

$$A_{vt} = \frac{R_p}{R_s + R_p}$$

Find F of terms of R_s and R_p

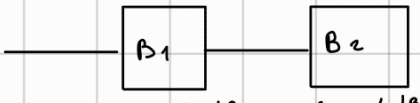
Gürültü masörüsü

Ex:



$$F = \frac{v^2_{n,out}}{A^2_{vt}} \qquad \frac{1}{4kTR_S} = ?$$

Ex



$G_1 = 10dB$ $\quad G_2 = 4dB$

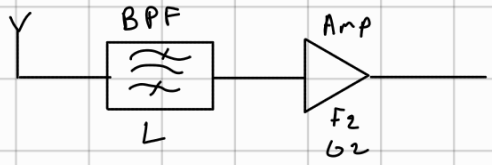
$NF_1 = 2dB$ $\quad NF_2 = 10dB$

$NF_t = ?$

Noise in Passive Reciprocal Circuit

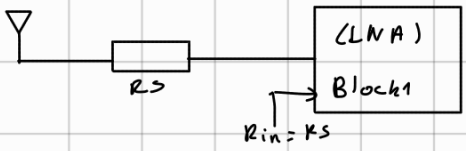
Eksempel What is reciprocity

Ex:



$F_e = ?$

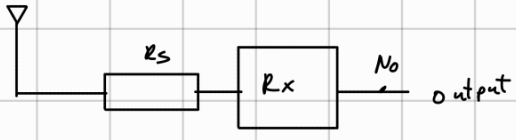
Noise power in Rx input



What is Pin? (Input noise power in bandwith B)

Receive Noise Floor

Noise floor = Receiver input equivalent noise

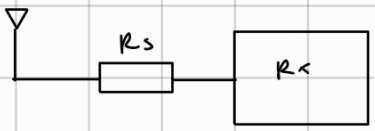


N_0: output noise power exluding R_S.

What is receiver Noise floor

Ders 7

Ex:



G = 16 dB
NF = 10 dB
B = 10 MHz

What is the noise floor = ?

What is $N_{out,t}$ = ?!

Ex:



NF = 3 dB
G = 12 dB

Calculate NF total and Noise floor (B = 5MHz) için ?

Sensitivity and Dynamic Range

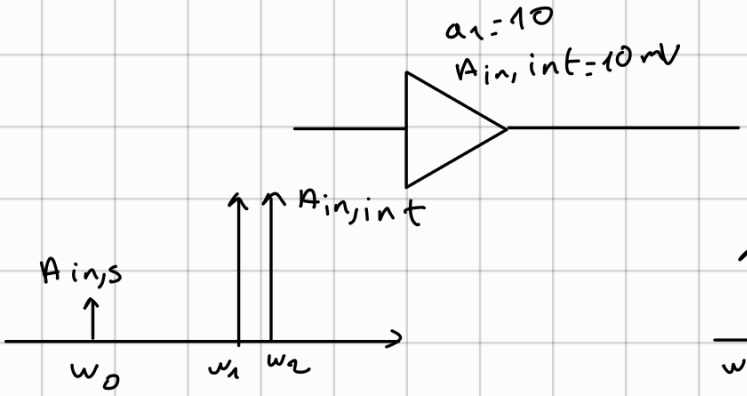
Ex: A GSM receiver has $SNR_{out(min)}$ = 12dB channel BW = 20khz

NF = 9 dB

Ekstra: What is the Dynamic Range and Spur free Dynamic Range formula

Ders 8

Ex:



$A_{IIP3} = 80\,mV$

→ Find $A_0(IM3) = ?$

$IM3 = ?$



Eksoru: List all the modulation Concepts

Ders 9 RF link

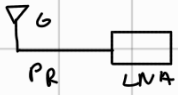
Ex: point-point (LOS)

$G_R = 60$ dB

$f = 60$ GHz

$P_{R\,density} = 1\,\mu W/cm^2$

what is the power at output of antenna connected to RF electronics



Ex: $f = 5.6$ GHz

$G_A = 35$ dB

$G_R = 6$ dB

$d = 200$ m

$\frac{1}{d^3} \rightarrow n = 3$

$L_{link} = ??$

$P_R = ?$ if $P_T = 20$ dBm ← ek soru

Link margin nasıl hesaplanır

Ders 10

Ex: $\eta_T = 0,5$ → microstrip antenna efficiency

$G_T = 11dB$ → antenna gain

$P_{in} = 40W$ (input power before antenna)

T_x power density falls as $\left(\frac{1}{d^2}\right)$

a) $P_{in} = _____ ??$ [dBm]

b) $P_T = ?$ [dBm]

c) $d = 5km \rightarrow P_D = ?$

d) $A_R = 6cm^2 \rightarrow P_R = ?$

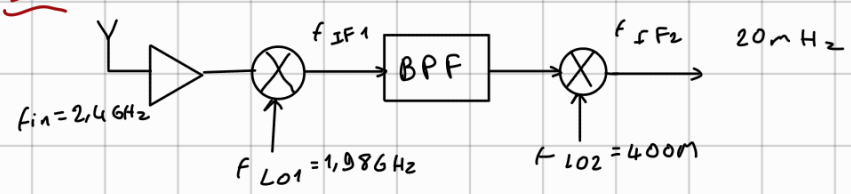
e) Alıcıdaki gürültü

Ek soru: Heterodyne ile Homodyne Receiver arasındaki fark nedir?

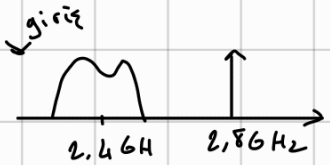
Ek soru: Down conversion işleminde High side injection ile low side injection farkı nedir?

Eksoru: Dual Conversion Receiver mimarisini çiziniz, image veinterference nasıl gokettiğini inceleginiz Amiçilerin lineerliği nasıl olmalıdır

Ex:



fLO2 (2^n harmonic) = 800MHz olduğuna göre karışım olur mu



Eksoru : Dual Conversion receivers avantaj ve dezavantajları nelerdir?

Eksoru IQ mixer blok diagramı çiz?

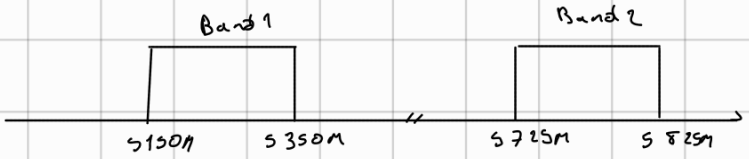
Sliding IF Receiver

Ex: $f_{LO2} = \frac{1}{4} f_{LO1}$

$|f_{in} - f_{LO1} - f_{LO2}| = 0$

Gelişme sorusu

Ex: 802.11a, ÷8 structure sliding Receiver



1) Image Band?

2) Mixing spurs in results of mixing with 3rd harmonic

& 3rd harmonic of second LO

$f_{in} - m f_{LO1} - f_{LO2} = 0$

$f_{in} - f_{LO1} - n f_{LO2} = 0$

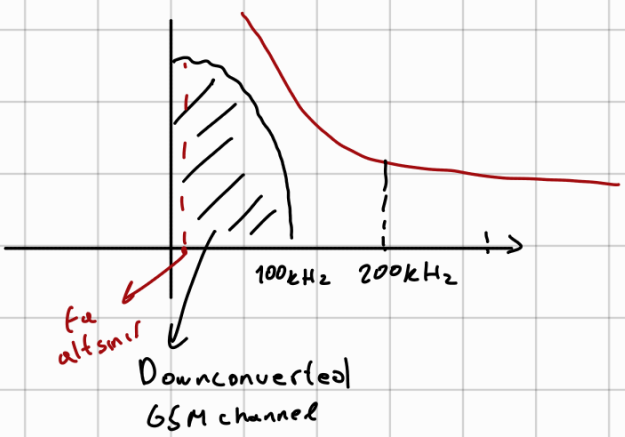
Eksoru: What is Zero IF direct conversion Receiver advantages

Eksoru: Draw the FSK Direct Conversion receiver

Ek soru What is Drawbacks of Direct Conversion Receiver

Soru:

2. Doğrudan aşağı çevirme işlemine sahip bir GSM alıcısına, 200kHz'de köşe frekansı olan "flicker" gürültüsü etki etmektedir. Temel band GSM işaretin band genişliği 100kHz'dir. "flicker" gürültüsü etkisinin 27Hz'den başladığı varsayılmaktadır.
   a) "Flicker" gürültüsü nedeniyle oluşan "flicker noise penalty" değerini hesaplayınız.
   b) Gürültü spektral yoğunluğunu hangi kazanç değeri kadar arttırılmalı ki (veya almaç kazancı ne kadar arttırılmalı ki) "flicker noise penalty" değeri 10 dB'nin altına düşsün?
   c) Bu durumda "flicker" gürültüsü köşe frekansı ne olmalıdır?



IQ Mismatch

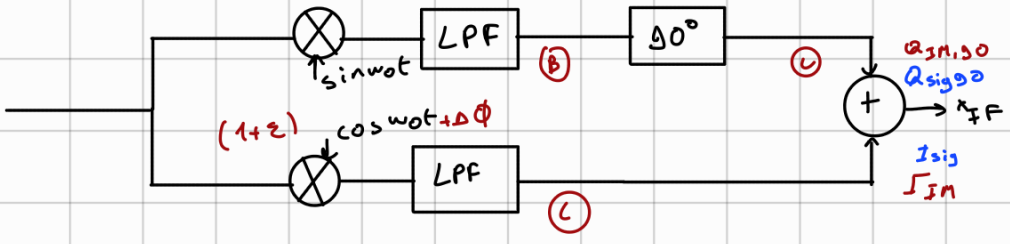
Ders13

Eksoru What is Hilbert Transform

Eksoru List the implementation of 90° phase shift

Eksoru Draw an Image Reject Receiver and explain it

Çalışma sorusu



$$x_{sig,out}(t) = \frac{A_{sig}}{2}\cos[(\omega_{RF}-\omega_{LO})t+\phi_{sig}] + \frac{A_{sig}}{2}(1+\varepsilon)\cos[(\omega_{RF}-\omega_{LO})t+\phi_{sig}+\Delta\theta]$$

$$x_{in,out}(t) = \frac{A_{in}}{2}(1+\varepsilon)\cos[(\omega_{IM}-\omega_{LO})t+\phi_{im}+\Delta\theta] + \frac{A_{in}}{2}\cos[(\omega_{LO}-\omega_{IM})t+\phi_{IM}]$$

Bilgi

Measure of Image Rejection

IRR : Image Rejection Ratio

$$\left.\frac{P_{im}}{P_{sig}}\right|_{out} = \frac{A_{im}^2}{A_{sig}^2} \cdot \frac{(1+\varepsilon)^2 - 2(1+\varepsilon)\cos\Delta\theta + 1}{(1+\varepsilon)^2 + 2(1+\varepsilon)\cdot\cos\Delta\theta + 1}$$

$A_{im} = A_{sig} \rightarrow$

for $\Delta\theta \ll 1$, $\varepsilon \ll 1$ $\rightarrow$ $IRR = \frac{4}{\varepsilon^2 + \Delta\theta^2}$

Soru

We have signal of -85dBm at 2,4 GHz and -10dBm at 5,2 GHz a interfere

What is the amount of rejection in BPF1 at 5,2 GHz for SNR = 20dB & IRR = 45dB

